



PTO Driveshaft Rust Protection

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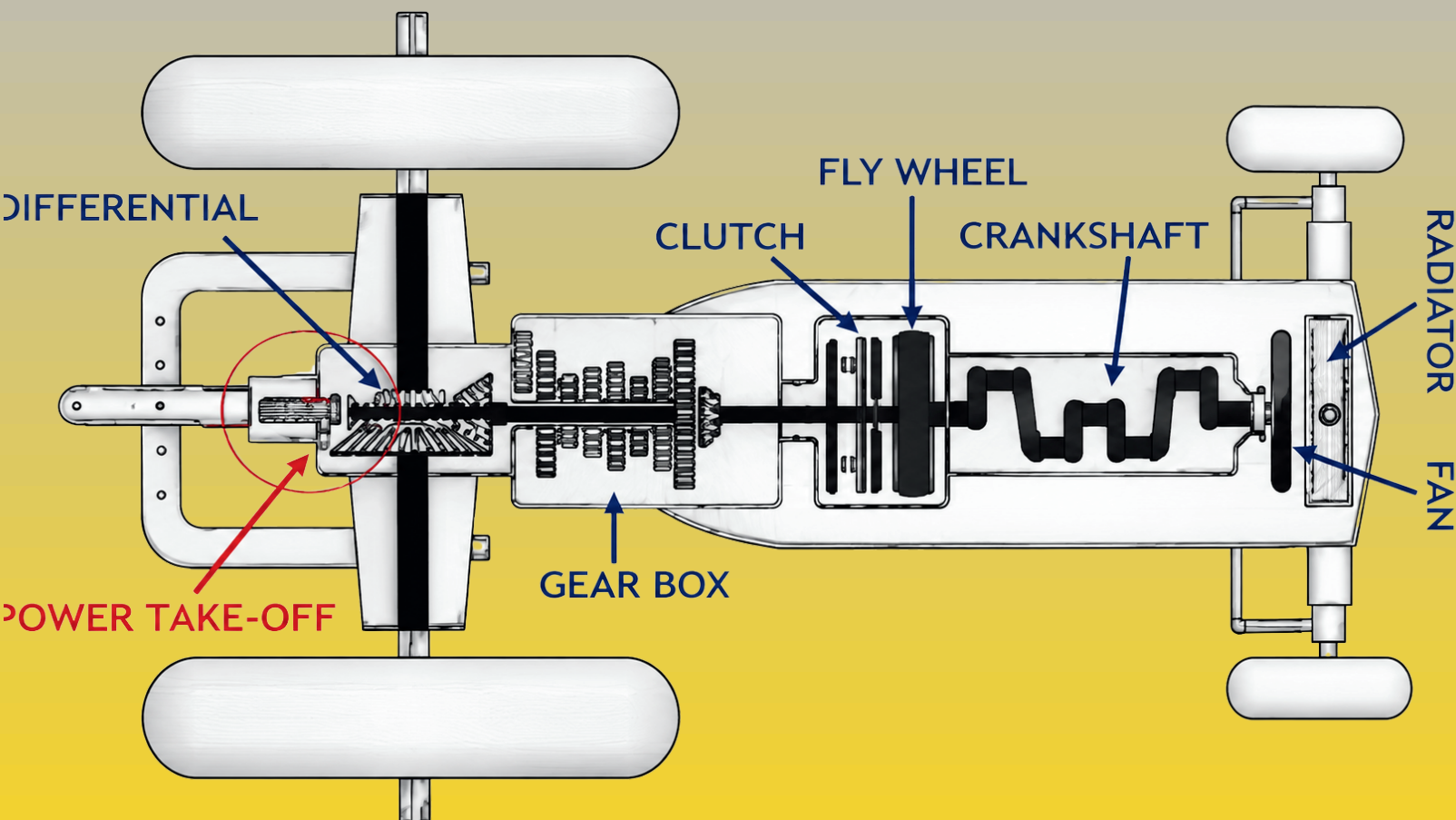


Background Info

PTO (Power Take-Off) Driveshafts are used to connect farming implements to the back of a tractor. These shafts often get water in them when they are detached and not being used. The water causes the iron to rust, causing difficulties in attaching the PTO shaft, and also causes a safety hazard while the shaft is in use.

Hypothesis

If something is inserted into the PTO driveshaft socket, it will block water and dirt from entering and will prevent rust.



Procedure

The procedure was to design a plug to fit into the PTO driveshaft for a tractor to prevent water and dirt from entering. A 3D-Printed plug was designed and re-iterated until it arrived on the final product.

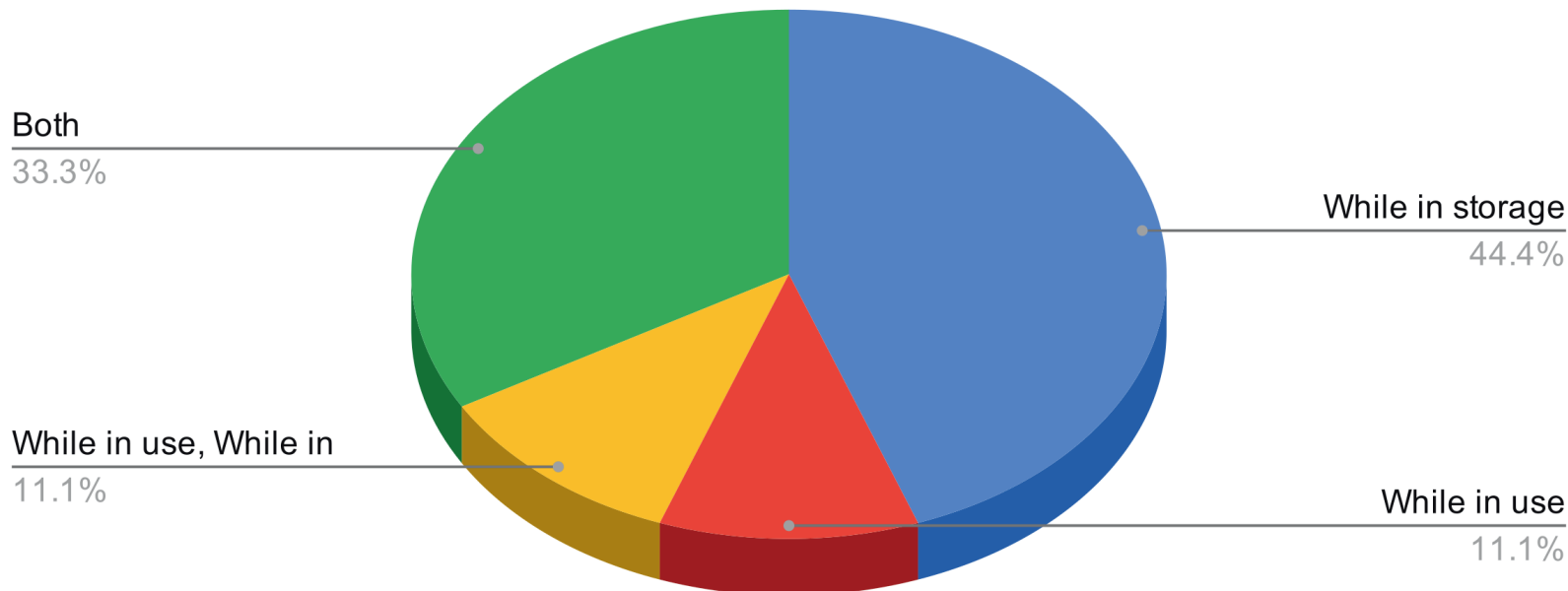
For works cited and more information, Scan the QR code below.



<https://rileyhagel.github.io/rtx/resources.htm>

The times people would like to use a solution based on a survey:

When would people use the solution? (While in Use or While in Storage)



Next Steps

The next steps are to find a way to mass-produce the plug after improving on the design. It could use injection molding instead of 3D printing, but it should be made of a rubbery material.

Baseline Data

0 Days	Start of Test	10/26/25
5 Days	Rust starts to form	11/1/25
10 Days	Noticeable rust	11/11/25
30 Days	Starts becoming hard to conect to the PTO because of rust	12/19/25
100 Days	Driveshaft is so corroded that it needs treatment and grease in order to connect	3/19/26

Goal	Want/Need	Met?
Easy to Use	Want	Yes
Stops Water and Dirt	Need	Yes
Can be Used while the shaft is being used	Want	No

Conclusion

The conclusion is that the driveshaft socket was well-protected from water and dirt, and it even retained its shape when it was run over by a tractor.

